RAHIAN ISLAM

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EDUCATION AND TRAINING

Schulich School of Engineering, University of Calgary - Calgary, AB *12/2024*

WORK EXPERIENCE

Mechanical Engineering EIT / Tarco Consulting Inc. - Calgary, AB *03/2025 – Present*

- Supported multi-disciplinary oil & gas projects for major clients such as TC Energy, Enbridge etc. including compressor station upgrades, valve replacements, glycol heating system upgrades, and piping/tubing modification scopes.
- Interpreted and applied ASME standards (B31.3, B16.5, B36.10), CSA codes, API standards and material specifications (e.g., ASTM A333 Gr.6) to support engineering design, procurement, and construction.
- Conducted detailed equipment and piping design reviews, including pressure drop calculations, pump curve analysis, control valve assessments, and piping layout optimization.
- Developed and reviewed technical deliverables such as construction work packages (CWPs), technical scopes, demolition plans, valve datasheets, line lists, valve lists, tie-point lists and BOMs in compliance with industry and client standards.
- Utilized Bluebeam and AutoCAD to produce and review P&IDs, piping GA layouts, isometrics, and equipment isolation plans to support safe construction and commissioning.
- Coordinated closely with drafting, QA/QC, and client representatives to incorporate markups, resolve design conflicts, and track progress across IFR and IFC package stages.
- Redrafted and standardized compressor technical specifications for Enbridge, consolidating multiple legacy documents into enforceable engineering standards and ensuring alignment with regulatory, HSE, and emissions compliance requirements.

Turbomachinery Fleet Management Intern / TC Energy - Calgary, AB *02/2024 – 12/2024*

- Dry Gas Seal (DGS) Failure Analysis – Analyzed historical DGS failures and categorized failure modes. Leveraged advanced Power BI tools to visualize failure trends that aided maintenance and reliability management decisions.
- Dry Gas Seal (DGS) Asset List- Developed a comprehensive DGS asset list (drive-end, non-drive-end, and barrier seals) by consolidating OEM/vendor data, drawings, part numbers, and SAP equipment numbers to improve asset management and traceability.
- Compressor Health Assessments (CHA) – Performed Compressor Health Assessments (CHA) using RAMP, DSS, and Fleet Manager data; authored reports evaluating mechanical condition, overhaul timing, and commercial impact to support overhaul vs. deferment decisions.

Compression, Civil and Mechanical Engineering Intern / TC Energy - Calgary, AB *05/2022 – 02/2024*

- Masterplan Tool – Led data migration of 3,000+ projects (compression, civil, mechanical) into the new Facilities Masterplan tool, improving visibility and planning of maintenance activities across Canada Gas Operations facilities.
- RAIL (Rolling Action Item Catalog) – Built and managed Excel Query interface connected to the Masterplan project database, enabling all PMs and other stakeholders to pull real-time project data and log notes for improved tracking and coordination.
- FMEA (Failure Mode and Effects Analysis) – Conducted FMEA by extracting corrective work orders from SAP, identifying failure modes and root causes, and visualizing trends in Power BI for critical systems (recycle valves, instrument air, aerial coolers).
- Technical Re-Aero Memo – Performed Balance of Plant (BOP) evaluations and authored Re-Aero technical memos, assessing piping, PSVs, recycle valves, scrubbers, and coolers to ensure safe operation under new compressor wheel configurations and initiated MOCs where required.
- Fuel Gas Meters and Regulators Sizing– Performed flow analysis and pressure drop calculations to size fuel gas regulators and meters, collaborating with senior engineers to ensure optimal performance and compliance with safety standards.
- Hydrotesting Reviews – Reviewed hydrotest packages for new facility construction, ensuring accurate test boundaries, eliminating redundant tests, and confirming piping integrity and readiness under P.Eng. supervision

TECHNICAL SKILLS

- AutoCAD
- MATLAB
- Microsoft Word, Excel, PowerPoint
- Isograph & Raptor – Equipment Sparing Analysis tool
- Python
- Power BI
- Bluebeam
- SAP (System Analysis Program)